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(/learn/data-manipulation/home/welcome)

**Bill Howe**

The MapReduce programming model (as distinct from its implementations) was proposed as a simplifying abstraction for parallel manipulation of massive datasets, and remains an important concept to know when using and evaluating modern big data platforms.

MAPREDUCE AND PARALLEL DATAFLOW PROGRAMMING

Lesson 10: Reasoning about Scale



What Does Scalable Mean? 6 min

(/learn/data-manipulation/lecture/UxKFs/what-does-scalable-mean)



A Sketch of Algorithmic Complexity 5 min

(/learn/data-manipulation/lecture/oqEYT/a-sketch-of-algorithmic-complexity)



A Sketch of Data-Parallel Algorithms 5 min

(/learn/data-manipulation/lecture/o7zNa/a-sketch-of-data-parallel-algorithms)



"Pleasingly Parallel" Algorithms 4 min

(/learn/data-manipulation/lecture/badi8/pleasingly-parallel-algorithms)



More General Distributed Algorithms 4 min

Help Center

(/learn/data-manipulation/lecture/9Yhv7/more-general-distributed-algorithms)

Lesson 11: The MapReduce Programming Model

 MapReduce Abstraction 5 min

(/learn/data-manipulation/lecture/1Fvee/mapreduce-abstraction)

 MapReduce Data Model 4 min

(/learn/data-manipulation/lecture/czgP4/mapreduce-data-model)

 Map and Reduce Functions 3 min

(/learn/data-manipulation/lecture/Lm1VY/map-and-reduce-functions)

 MapReduce Simple Example 4 min

(/learn/data-manipulation/lecture/LBLUh/mapreduce-simple-example)

✓ MapReduce Simple Example
cont'd 3 min

(/learn/data-manipulation/lecture/owjMR/mapreduce-simple-example-cont-d)

 MapReduce Example: Word Length Histogram 3 min

(/learn/data-manipulation/lecture/0V0Gu/mapreduce-example-word-length-histogram)

 MapReduce Examples: Inverted Index, Join 6 min

(/learn/data-manipulation/lecture/LzEBt/mapreduce-examples-inverted-index-join)

Lesson 12: Algorithms in MapReduce

 Relational Join: Map Phase 4 min

(/learn/data-manipulation/lecture/khmOZ/relational-join-map-phase)

 Relational Join: Reduce Phase 5 min

(/learn/data-manipulation/lecture/NWgaE/relational-join-reduce-phase)

 Simple Social Network Analysis:
Counting Friends 4 min

(/learn/data-manipulation/lecture/B4aRJ/simple-social-network-analysis-counting-friends)

 Matrix Multiply Overview 6 min

(/learn/data-manipulation/lecture/SNJY5/matrix-multiply-overview)

 Matrix Multiply Illustrated 4 min

(/learn/data-manipulation/lecture/QW35z/matrix-multiply-illustrated)

 Shared Nothing Computing 5 min

(/learn/data-manipulation/lecture/Ugdel/shared-nothing-computing)

 MapReduce Implementation 5 min

(/learn/data-manipulation/lecture/qTMZM/mapreduce-implementation)

 MapReduce Phases 6 min

(/learn/data-manipulation/lecture/Gyfm1/mapreduce-phases)

Lesson 13: Parallel Databases vs. MapReduce

 A Design Space for Large-Scale Data Systems 5 min

(/learn/data-manipulation/lecture/YDNBO/a-design-space-for-large-scale-data-systems)

 Parallel and Distributed Query Processing 6 min

(/learn/data-manipulation/lecture/1HIDA/parallel-and-distributed-query-processing)

 Teradata Example, MR Extensions 5 min

(/learn/data-manipulation/lecture/O24yB/teradata-example-mr-extensions)

 RDBMS vs. MapReduce: Features 7 min

(/learn/data-manipulation/lecture/h36q0/rdbms-vs-mapreduce-features)

 RDBMS vs. Hadoop: Grep 6 min

(/learn/data-manipulation/lecture/2WKzp/rdbms-vs-hadoop-grep)

 RDBMS vs. Hadoop: Select,



Aggregate, Join 3 min

(/learn/data-manipulation/lecture/nfUMj/rdbms-vs-hadoop-select-aggregate-join)

Assignment 3: MapReduce



Assignment: Thinking in
MapReduce 3h 00m

(/learn/data-manipulation/programming/Dp7qI/thinking-in-mapreduce)